Universität Hamburg
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Universitätsklinikum Hamburg-Eppendorf

Predicting Protein Crystallization Conditions

Data Acquisition, Preprocessing, Exploratory Data Analysis

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Maschinelles Lernen in der Bioinformatik

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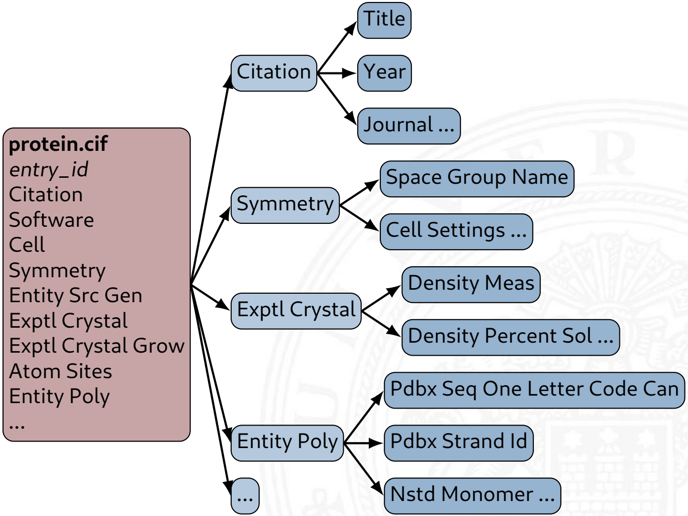
Outline

From monolithic CIF to relational tables

1. Filter out entries not crystallized using X-Ray Crystallography 10%
2. Filter out entries which did not have any crystallization information
3. Restructure database from monolithic to relational



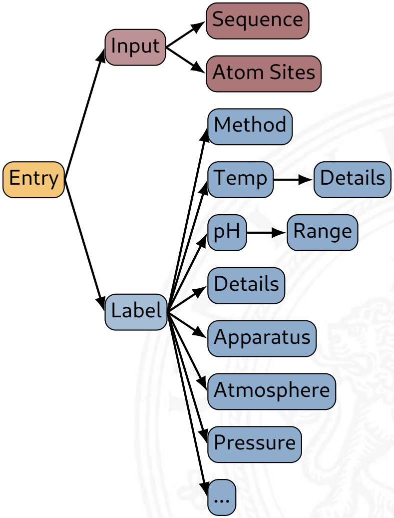
Data Description

Number of entries: 196743
Number of features: 612

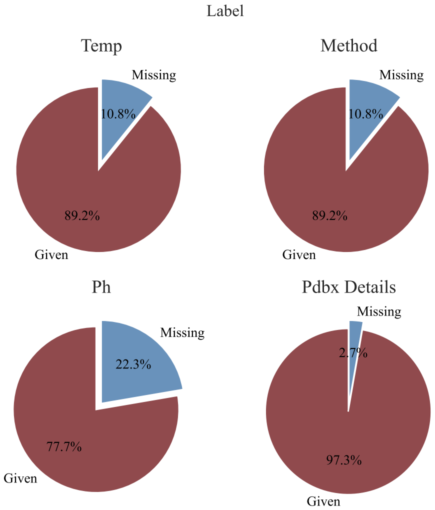
Features related to input : 2

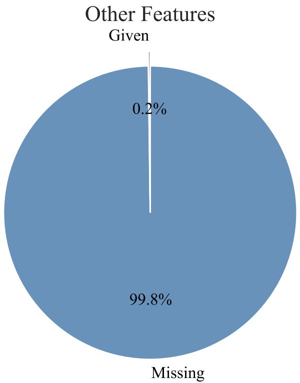
Features related to label : 18

BUT!

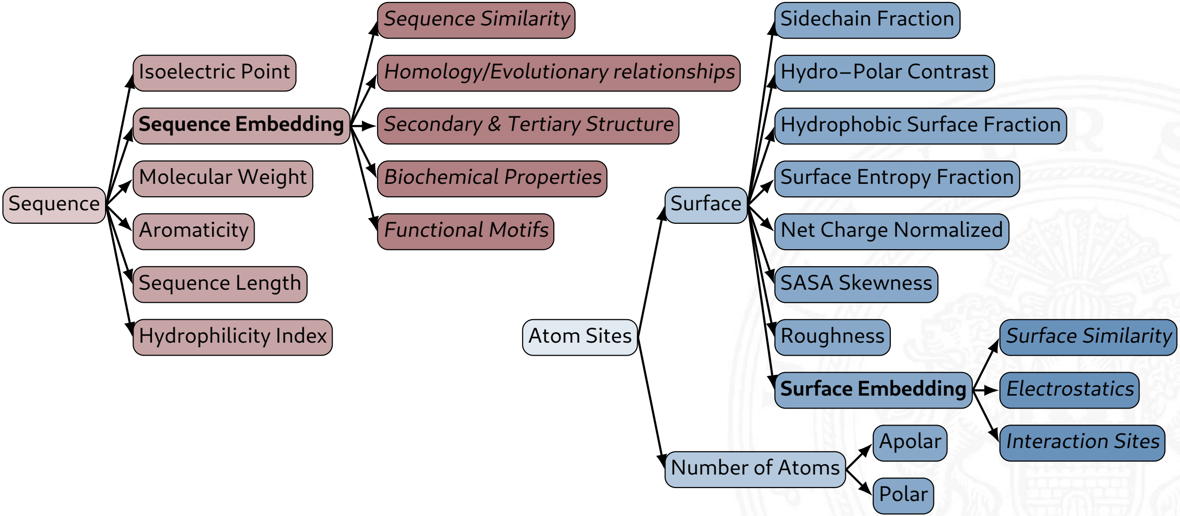


Missing Label





Derivative Features

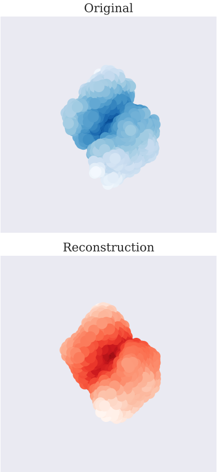


Sequence Embedding

- ESM based from Rives et al., 2021
- trained on 250m protein sequences
- biochemical properties of amino acids to remote homology of proteins
- deep Transformer learned using masked language modeling objective

Surface Embedding

- trained on 200k protein structures
- surface sampled using dMaSif (Sverrisson et al., 2020)
- Auto-Encoder approach to create surface embedding



Embeddings

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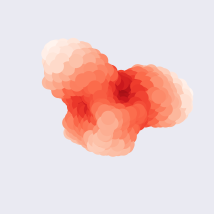
Original

Reconstruction



Embeddings

Sequence Embedding

- ESM based from Rives et al., 2021
- trained on 250m protein sequences
- biochemical properties of amino acids to remote homology of proteins
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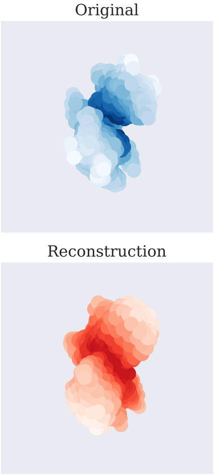
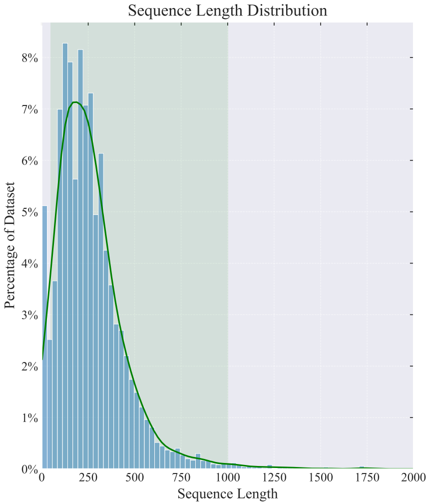


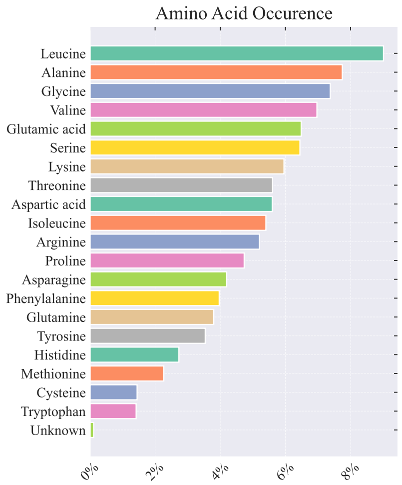


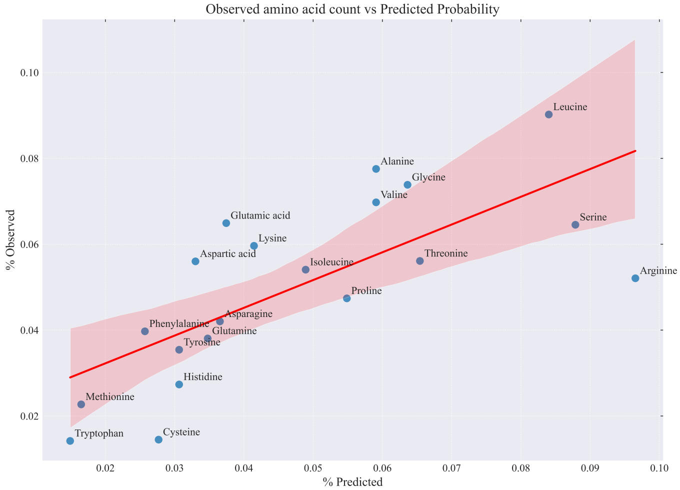
Sequence Length Distribution

- Input size has to be chosen such that the majority of the protein space is included in the dataset
- Outliers might unnecessarily increase model size or are too small to encode information (92% for [60, 1000])

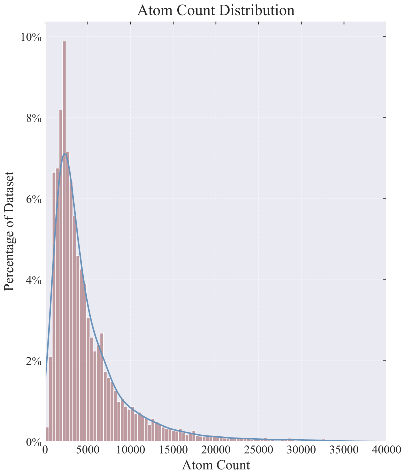
	Sequence Length
count	280543
mean	262
std	191
min	2
max	4128

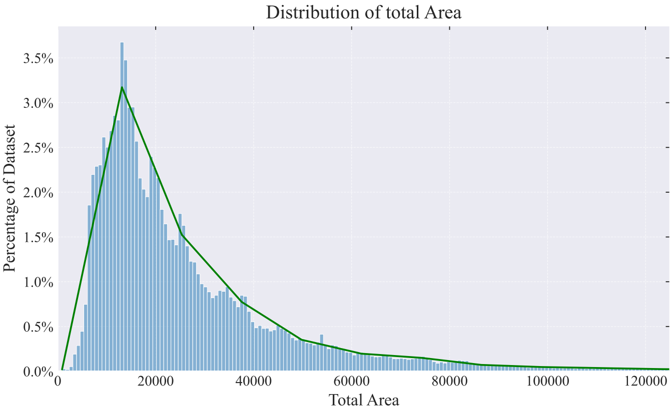
Amino Acid Distribution



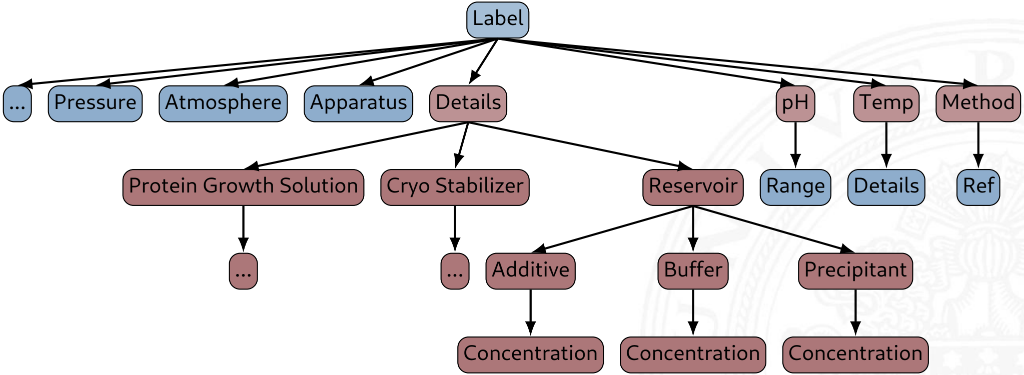


Atom Count Distribution

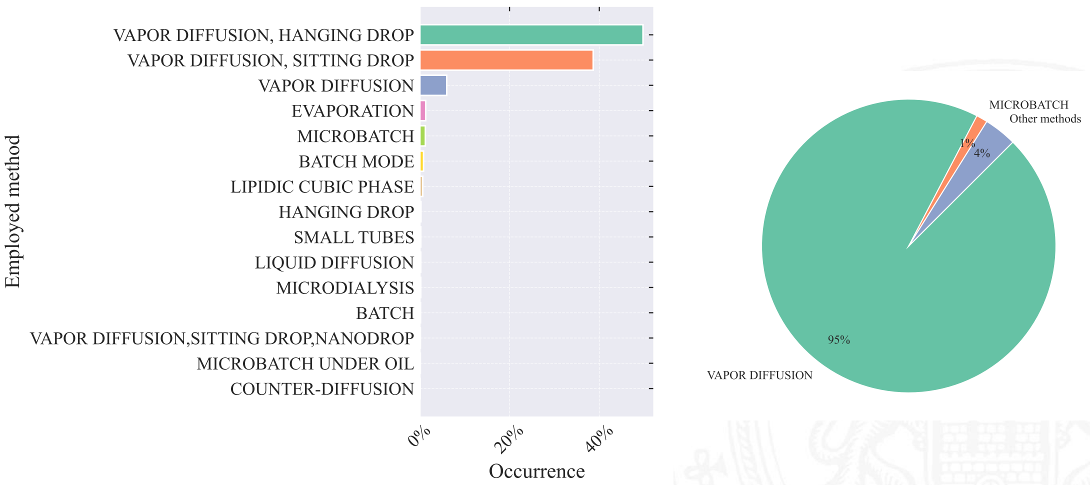




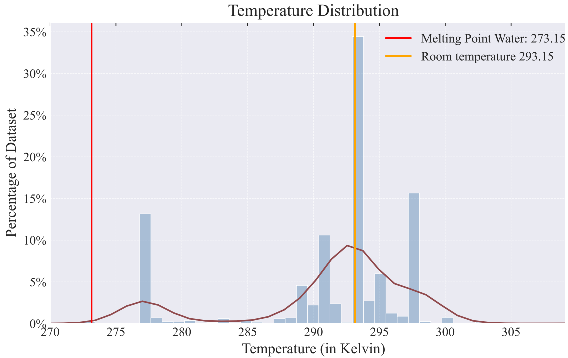
Label Features

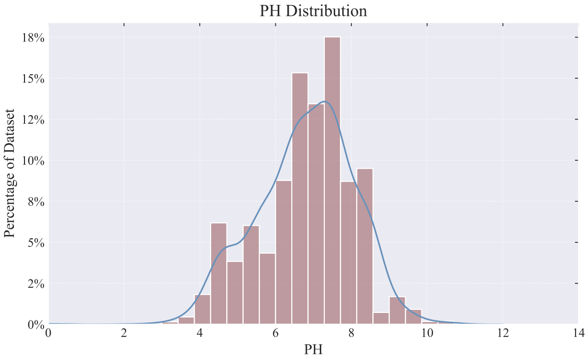


Method of Crystallization



Temperature and pH





Label Parsing (Uniform Words)

Based on preprocessing proposed by Lynch, Dudek, and Bowman, 2020

Crystallization was carried out in sitting-drop vapor-diffusion setups with 1:1 mixtures of protein solution containing 0.7 mM Cko and 1.8 mM MnCl2 and reservoir solution containing 19%-21% PEG 3350 and 30 **percent polyethylene glycol monomethyl ether** 4000 and 0.2 M Na2HPO4 , pH 9.5, VAPOR DIFFUSION, SITTING DROP, temperature 298K

↓

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Label Parsing (Split Sections)

Crystallization was carried out in sitting-drop vapor-diffusion setups with 1:1 mixtures of **protein solution** containing 0.7 mM Cko and 1.8 mM MnCl2 and **reservoir solution** 19%-21% PEG 3350 and 30% PEGMME 4000 and 0.2 M Na2HPO4 , **pH 9.5, VAPOR DIFFUSION, SITTING DROP, temperature 298K**

↓

protein solution : containing 0.7 mM Cko and 1.8 mM MnCl2 and
reservoir solution : containing 19%-21% PEG 3350 and 30% PEGMME 4000 and 0.2 M Na2HPO4
pH 9.5 ,
VAPOR DIFFUSION, SITTING DROP ,
temperature 298K

Label Parsing (Average Ranges)

protein solution: containing 0.7 mM Cko and 1.8 mM MnCl2 and
reservoir solution: containing **19%-21%** PEG 3350 and 30% PEGMME 4000 and 0.2 M Na2HPO4 ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

↓

protein solution: containing 0.7 mM Cko and 1.8 mM MnCl2 and
reservoir solution: containing **20%** PEG 3350 and 30% PEGMME 4000 and 0.2 M Na2HPO4 ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

Label Parsing (Normalize Units)

protein solution: containing 0.7 mM Cko and 1.8 mM MnCl2 and
reservoir solution: containing 20% PEG 3350 and 30% PEGMME 4000 and **0.2 M** Na2HPO4 ,
pH 9.5,
VAPOR DIFFUSION,
SITTING DROP, temperature 298K

↓

protein solution: containing 0.7 mM Cko and 1.8 mM MnCl2 and
reservoir solution: containing 20% PEG 3350 and 30% PEGMME 4000 and **200 mM** Na2HPO4 ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

Label Parsing (Remove unnecessary words)

protein **solution** : **containing** 0.7 mM Cko **and** 1.8 mM MnCl2 **and**
reservoir **solution** : **containing** 20% PEG 3350 **and** 30% PEGMME 4000 **and** 0.2 M Na2HPO4 ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

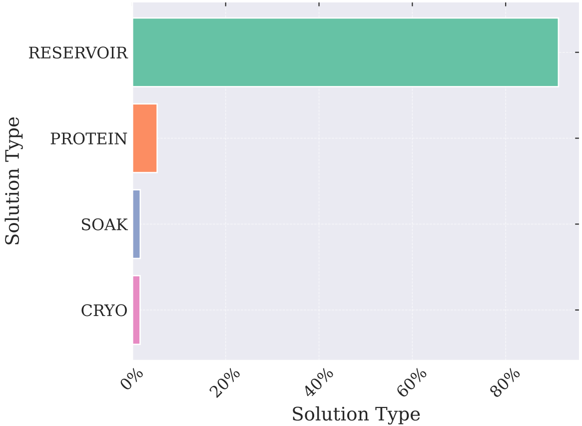
↓

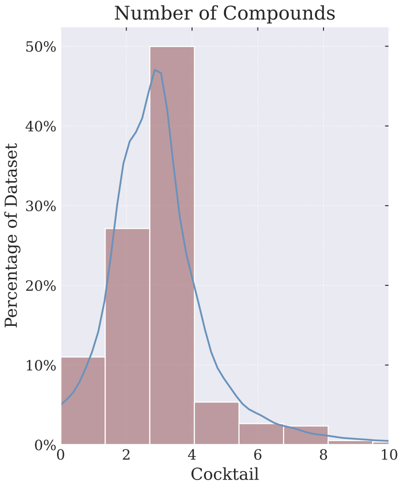
protein: 0.7 mM Cko 1.8 mM MnCl2
reservoir: 20% PEG 3350 30% PEGMME 4000 200 mM Na2HPO4 ,
pH 9.5,
VAPOR DIFFUSION, SITTING DROP,
temperature 298K

Label Parsing (Format)

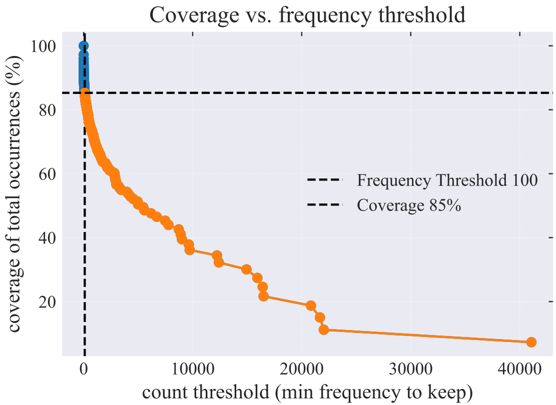
```
{
"protein": {
        "Cko": 0.7mM,
        "MnCl2": 1.8mM,

        },
"reservoir": {
        "PEG 3350": 20\%,
        "PEGMME 4000": 30\%,
        "Na2HPO4": 200mM,

        },
"pH": 9.5,
"method": Vapor Diffusion, Sitting Drop
"temperature": 298k
}
```

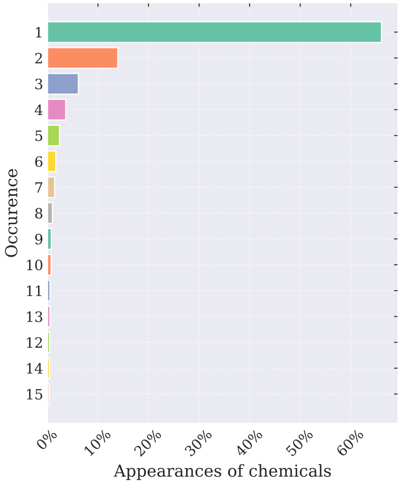
Compound Distribution



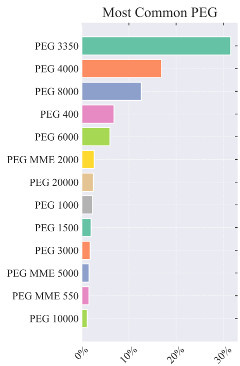


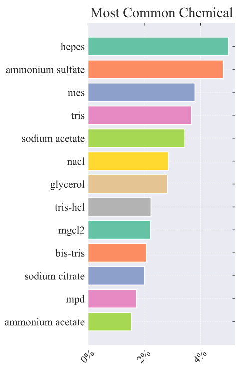
- Number of unique chemicals: 24999
- After frequency filtering at 100: 348

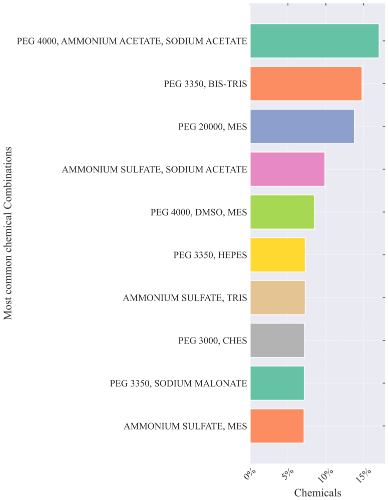




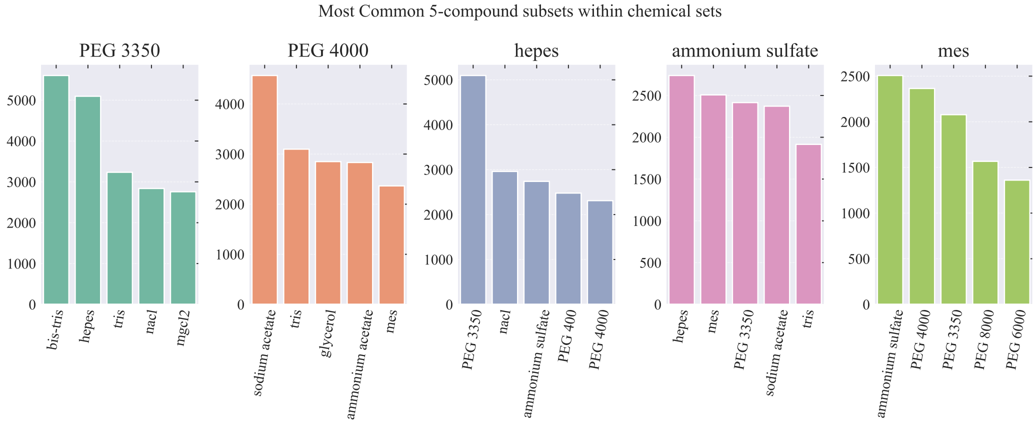
Compound Distribution



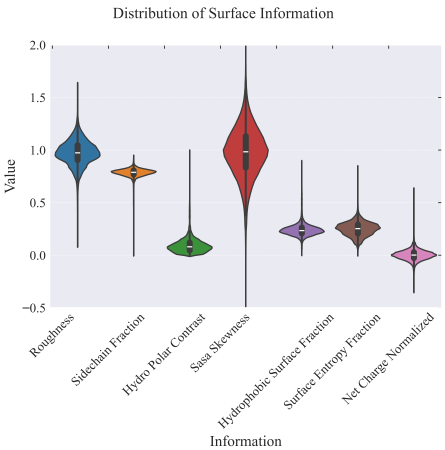


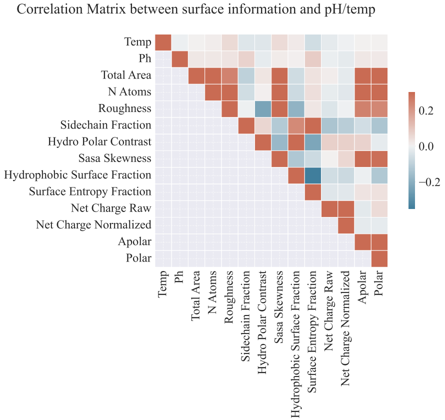


Chemical Cocktails

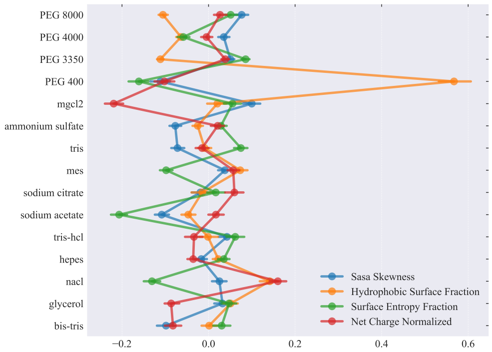


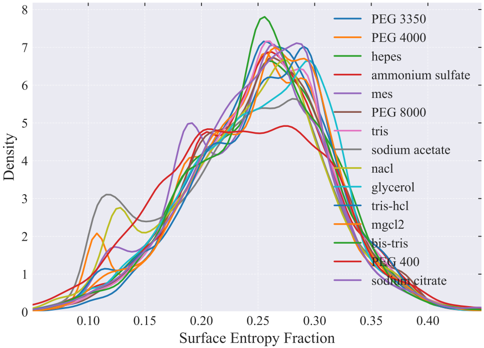
Surface Features vs. Ph/Temperature





Surface Features vs. Compounds





Crystal Features vs. Compounds

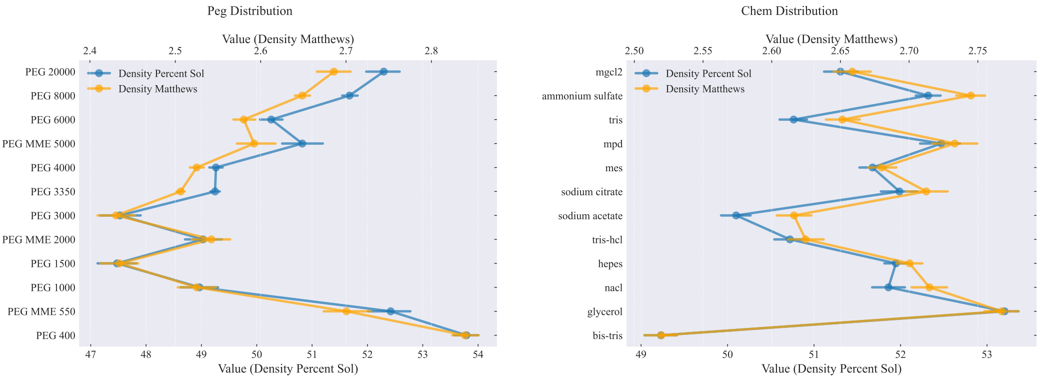
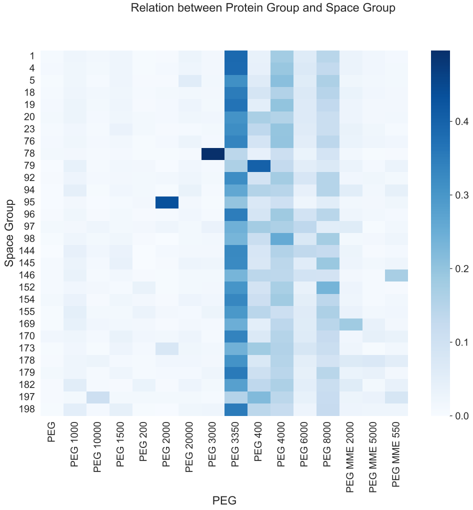
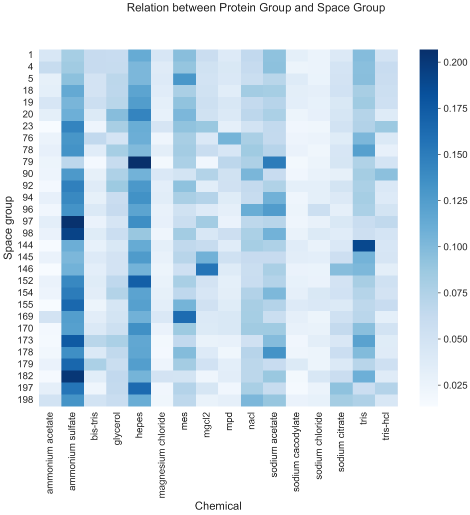


Figure: Correlation of Solvent Content and Compounds. Distributions: Slide: 40 & 41

Crystal Features vs. Compounds





Crystal Features vs. Compounds

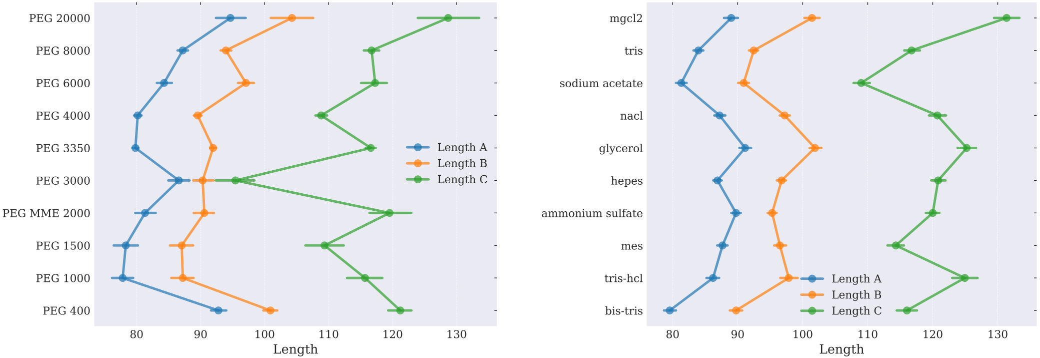


Figure: Relation of Crystal Dimensions and Compounds. Distributions: Slide: 42

Crystal Features vs. Compounds

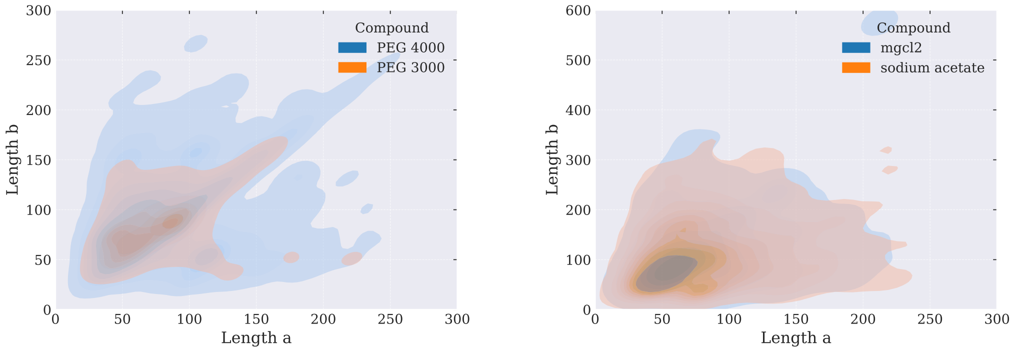
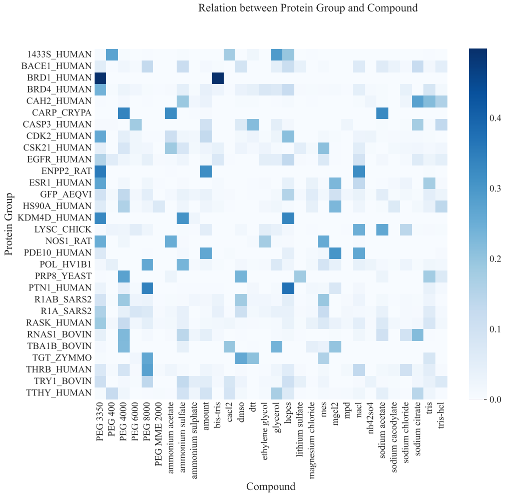
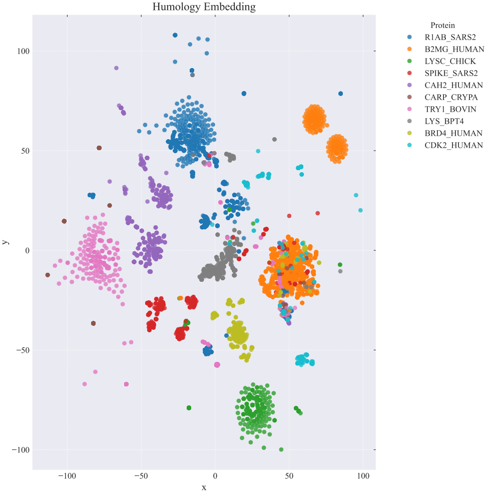
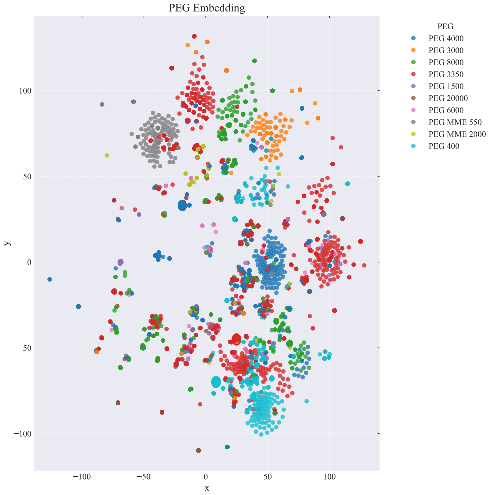


Figure: Relation of Crystal Dimensions and Compounds. Distributions: Slide: 42

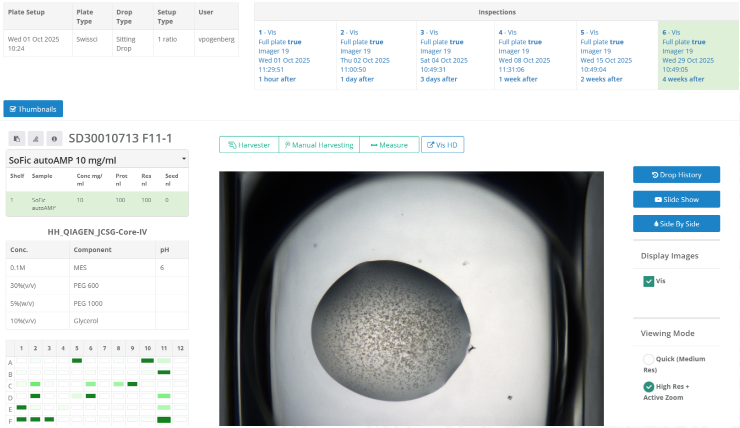

Relation between Protein Group and Compound
Protein Group
1433S_HUMAN
BACE1_HUMAN
BRD1_HUMAN
BRD4_HUMAN
CAH2_HUMAN
CARP_CRYPA
CASP3_HUMAN
CDK2_HUMAN
CSK21_HUMAN
EGFR_HUMAN
ENPP2_RAT
ESR1_HUMAN
GFP_AEQVI
HS90A_HUMAN
KDM4D_HUMAN
LYSC_CHICK
NOS1_RAT
PDE10_HUMAN
POL_HV1B1
PRP8_YEAST
PTN1_HUMAN
R1AB_SARS2
R1A_SARS2
RASK_HUMAN
RNAS1_BOVIN
TBA1B_BOVIN
TGT_ZYMMO
THRB_HUMAN
TRY1_BOVIN
TTHY_HUMAN
Compound
PEG 3350
PEG 400
PEG 4000
PEG 6000
PEG 8000
PEG MME 2000
ammonium acetate
ammonium sulfate
ammonium sulphate
amount
bis-tris
cacl2
dmso
dtt
ethylene glycol
glycerol
hepes
lithium sulfate
magnesium chloride
mes
mgcl2
mpd
nacl
nh42so4
sodium acetate
sodium cacodylate
sodium chloride
sodium citrate
tris
tris-hcl
0.4
0.3
0.2
0.1
0.0


Humology Embedding
y
100
50
0
−50
−100
x
−100
−50
0
50
100
Protein
R1AB_SARS2
B2MG_HUMAN
LYSC_CHICK
SPIKE_SARS2
CAH2_HUMAN
CARP_CRYPA
TRY1_BOVIN
LYS_BPT4
BRD4_HUMAN
CDK2_HUMAN

Compounds in Embedding



- Beamline Evaluation Platform
- each plate 96 conditions
- drops are monitored 1h, 1-3days, 1-4weeks, -4months
- some images are automatically rated
- delivers negative data



Rating

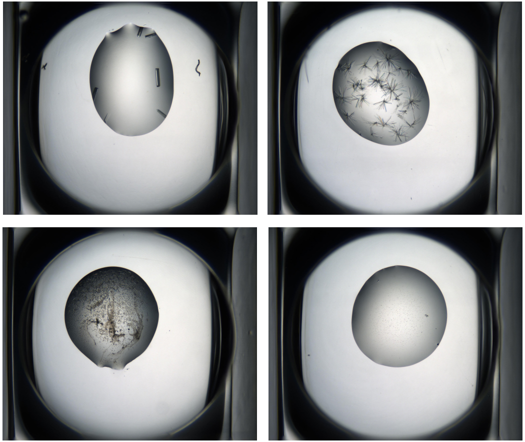
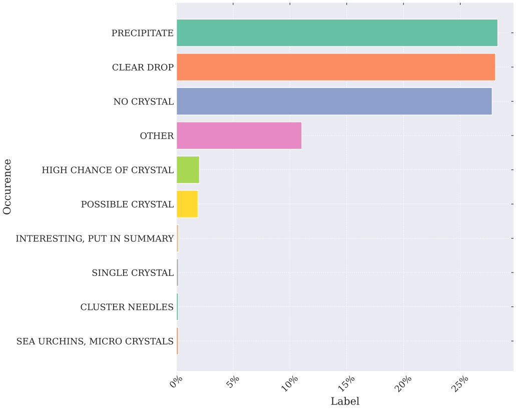
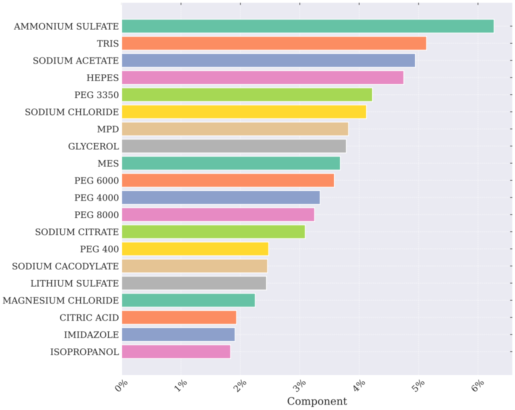
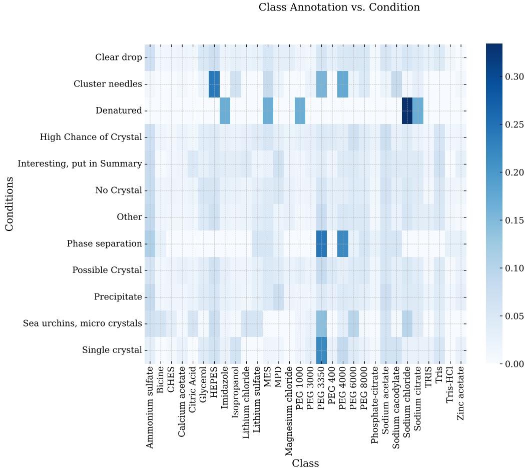


Figure: Scores: 0.99, 0.97, 0.90, 0.3



Conditions


AMMONIUM SULFATE
TRIS
SODIUM ACETATE
HEPES
PEG 3350
SODIUM CHLORIDE
MPD
GLYCEROL
MES
PEG 6000
PEG 4000
PEG 8000
SODIUM CITRATE
PEG 400
SODIUM CACODYLATE
LITHIUM SULFATE
MAGNESIUM CHLORIDE
CITRIC ACID
IMIDAZOLE
ISOPROPANOL
0%
1%
2%
3%
4%
5%
6%
Component


Class Annotation vs. Condition
Conditions
Clear drop
Cluster needles
Denatured
High Chance of Crystal
Interesting, put in Summary
No Crystal
Other
Phase separation
Possible Crystal
Precipitate
Sea urchins, micro crystals
Single crystal
Ammonium sulfate
Bicine
CHES
Calcium acetate
Citric Acid
Glycerol
HEPES
Imidazole
Isopropanol
Lithium chloride
Lithium sulfate
MES
MPD
Magnesium chloride
PEG 1000
PEG 3000
PEG 3350
PEG 400
PEG 4000
PEG 6000
PEG 8000
Phosphate-citrate
Sodium acetate
Sodium cacodylate
Sodium chloride
Sodium citrate
TRIS
Tris
Tris-HCl
Zinc acetate
Class
0.30
0.25
0.20
0.15
0.10
0.05
0.00

Summary & Hypothesis Generation

- while not strongly correlated surface features and embedding seem to have a predictive value of crystallization conditions
- literature indicates that crystal features have also a strong effect on conditions
- temperature/method space is limited focus more on pH and cocktail composition

Next Steps

- Pre-train on CRIMS data
- Task Design
- Model Design

References

Lynch, Miranda L., Max F. Dudek, and Sarah E. J. Bowman (July 2020). “A Searchable Database of Crystallization Cocktails in the PDB: Analyzing the Chemical Condition Space”. In: *Patterns* 1.4, p. 100024. ISSN: 2666-3899. DOI: 10.1016/j.patter.2020.100024. URL: https://www.sciencedirect.com/science/article/pii/S2666389920300246 (visited on 11/06/2025).

Rives, Alexander et al. (Apr. 2021). “Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences”. In: *Proceedings of the National Academy of Sciences* 118.15, e2016239118. DOI: 10.1073/pnas.2016239118. URL: https://www.pnas.org/doi/full/10.1073/pnas.2016239118 (visited on 11/06/2025).

Sverrisson, Freyr et al. (Dec. 2020). *Fast end-to-end learning on protein surfaces*. en. Tech. rep. Type: article. bioRxiv. Chap. New Results, p. 2020.12.28.424589. DOI: 10.1101/2020.12.28.424589. URL: https://www.biorxiv.org/content/10.1101/2020.12.28.424589v1 (visited on 11/06/2025).

Crystal Features vs. Compounds

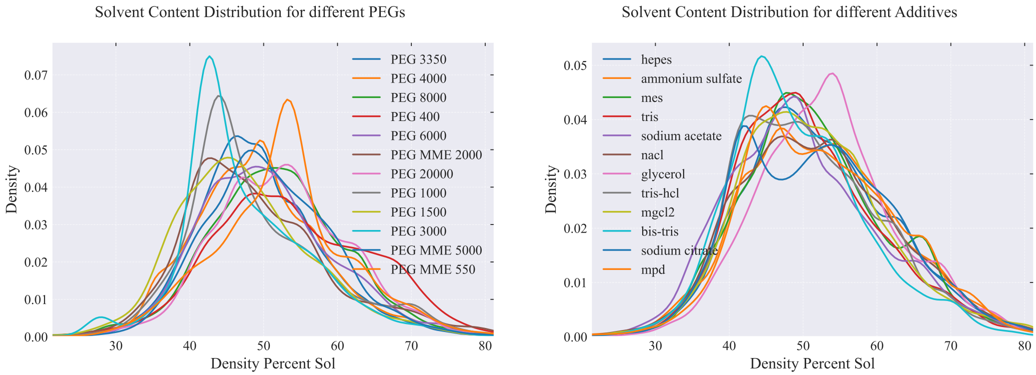


Figure: Correlation of Solvent Content and Compounds.

Crystal Features vs. Compounds

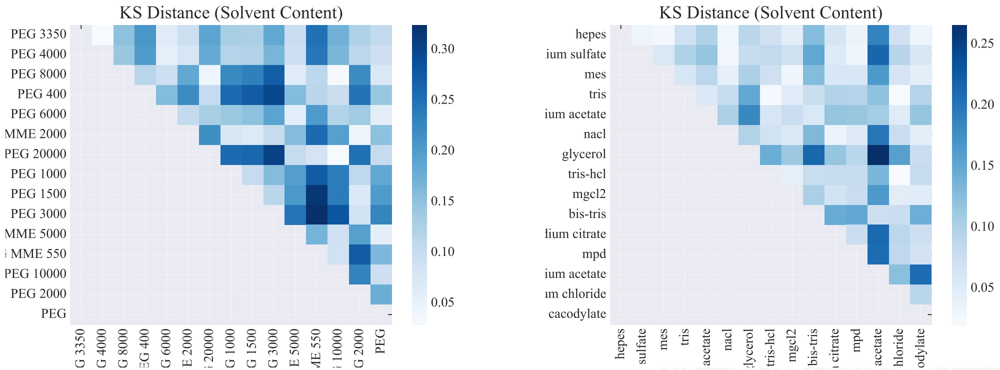


Figure: KS Distance between Compounds

Cell Features vs. Compounds